

The sharp square-root bill of an exact causal source

Full proof, audit bridge, and certified finite-horizon bounds

3 October 2026

Abstract

Let P, Q be finite probability laws of equal entropy h , with surprisal standard deviations $a \geq b \geq 0$. We prove

$$C_n^{\text{causal}}(P, Q) = nh + \sqrt{2/\pi} (a - b)\sqrt{n} + o(\sqrt{n})$$

for the full history-indexed exact causal model, including $b = 0$. The proof has two independent parts. A finite greedy response-tree decomposition realizes the Bellman information envelope within $\log_2 e$ bits at every horizon. An explicit two-Gaussian profile approximates the centered Bellman recursion; Taylor's theorem and sup-norm nonexpansiveness give a finite-horizon comparison with explicit constants. Gaussian tail bounds then give convergence of the means. No simultaneous coupling of policy diffusion limits, stochastic differential equation convergence, or unproved recycling lemma is used. For equal positive varentropies, a separate application of a published nonuniform martingale Berry–Esseen bound yields an $O(\log(n+1))$ excess.

1 The statement and the exact causal model

Zero-probability symbols are removed. Entropies and surprisals are in bits. Let

$$H(P) = H(Q) = h, \quad a^2 = \text{Var}_P(-\log_2 P(X)), \quad b^2 = \text{Var}_Q(-\log_2 Q(Y)), \quad a \geq b.$$

A seed S is sampled once, independently of external policy randomness. At round t , its output is a deterministic function of S , the complete past action-output history, and the currently requested action. Under every adaptive policy, its conditional output law must be the requested P or Q . Let C_n^{causal} be the infimum entropy of finite seeds with this property. Seeds may depend on n . Responses at distinct histories are not required to be identical.

Theorem 1. *For any such pair of finite laws,*

$$C_n^{\text{causal}}(P, Q) = nh + \sqrt{\frac{2}{\pi}} (a - b)\sqrt{n} + o(\sqrt{n}). \quad (1)$$

If $a = b = 0$, then $C_n^{\text{causal}} = nh$ exactly. If $a = b > 0$, then also

$$0 \leq C_n^{\text{causal}} - nh = O(\log(n+1)). \quad (2)$$

Write ν_P, ν_Q for the laws of the centered one-letter surprisals $D_A = -\log_2 P_A(Y) - h$, so $\mathbb{E}D_A = 0$. Define

$$(Tf)(x) = \min_{A \in \{P, Q\}} \mathbb{E}f(x - D_A), \quad G_0(x) = \mathbf{1}_{\{x \geq 0\}}, \quad G_n = T^n G_0. \quad (3)$$

Every G_n is a finite-support CDF; let $Z_n \sim G_n$ and $e_n = \mathbb{E}Z_n$. For a deterministic policy π , let $M_{\pi,n}$ be its transcript surprisal minus nh . Backward induction gives the exact identity

$$G_n(x) = \min_{\pi} \mathbb{P}(M_{\pi,n} \leq x). \quad (4)$$

Indeed, after choosing the first action and observing its output, each continuation policy can be selected separately for the remaining threshold. This is precisely (3). Allowing policy randomization does not reduce the minimum: conditioning on its randomness expresses its success probability as a mixture of deterministic-policy values.

The two ingredients of the proof are

$$0 \leq C_n^{\text{causal}} - nh - e_n \leq \log_2 e, \quad \frac{e_n}{\sqrt{n}} \rightarrow \sqrt{2/\pi} (a - b). \quad (5)$$

Both will be proved below.

2 Global realization with a constant entropy gap

We prove a finite-tree statement, also valid for history-dependent kernels and any finite number of actions. A history is v , and its child after action A and output y is vAy . Set

$$p(\emptyset) = 1, \quad p(vAy) = p(v)K_v(y \mid A).$$

These cylinder weights satisfy $\sum_y p(vAy) = p(v)$ for each action separately. For a policy π , let $\mathcal{L}_\pi$ denote its terminal histories. Their weights sum to one and form its transcript law. Define $I_\pi = -\log_2 p(T_\pi)$ and

$$F_*(t) = \min_{\pi} \mathbb{P}(I_\pi \leq t), \quad Z \sim F_*, \quad L = \mathbb{E}Z.$$

Lemma 2 (Finite global realization). *There is a finite exact causal seed S_G , supported on at most the number of terminal histories in the full tree, such that*

$$L \leq C \leq H(S_G) \leq L + \log_2 e. \quad (6)$$

Proof. A residual flow r is nonnegative and satisfies $\sum_y r(vAy) = r(v)$ for every action. A deterministic response strategy d chooses an output at each history-action pair. Its compatibility indicator $\chi_d(v)$ equals one when all its outputs on the path to v agree with v . It is a flow of root mass one.

For any residual flow compute from the leaves upward

$$B_r(v) = r(v) \quad (v \text{ terminal}), \quad B_r(v) = \min_A \max_y B_r(vAy) \quad (v \text{ nonterminal}).$$

There are two elementary consequences. First, the largest extractable strategy weight is $B_r(\emptyset)$: for each possible action the strategy selects a child of maximal bottleneck, and this is done recursively. Nonnegativity on terminal histories implies nonnegativity at all histories by flow conservation. Second, a policy choosing a minimizing action at every history has

$$r(v) \leq B_r(\emptyset) \quad (v \in \mathcal{L}_{\pi_r}).$$

Indeed, along every output of that policy, bottleneck values never increase. This policy also proves that no larger strategy weight is possible. If the root mass is positive, every action has a positive child, and induction gives a positive root bottleneck.

Starting from $r = p$, repeatedly extract a maximizing strategy of weight $w = B_r(\emptyset)$ and replace r by $r - w\chi_d$. At least one positive terminal residual becomes zero at each extraction; otherwise the same strategy would permit a larger weight. Residuals never increase, so the

algorithm terminates after finitely many steps. Its weights w_j are nonincreasing, sum to one, and satisfy

$$p(v) = \sum_j w_j \chi_{d_j}(v). \quad (7)$$

Choosing strategy d_j with probability w_j gives all the prescribed history weights simultaneously. Ratios of child to parent weights give the correct conditional laws at every positive-probability history. Thus this is one exact causal seed for all policies.

Fix $0 < \delta \leq 1$ and stop just after all weights larger than δ have been extracted. The remaining root mass is $r(\emptyset) = \sum_{j:w_j \leq \delta} w_j$. Its bottleneck is at most δ . The minimizing policy above therefore gives

$$\begin{aligned} \sum_{j:w_j \leq \delta} w_j &\leq \max_{\pi} \sum_{v \in \mathcal{L}_{\pi}} \min\{p(v), \delta\} \\ &= \max_{\pi} \mathbb{E} \min\{1, \delta 2^{I_{\pi}}\} \leq \mathbb{E} \min\{1, \delta 2^Z\}. \end{aligned} \quad (8)$$

The last inequality uses stochastic dominance: $F_* \leq F_{I_{\pi}}$. The zero residual case satisfies the same inequality trivially. For $J = -\log_2 w_{S_G}$, substitute $\delta = 2^{-t}$ to obtain

$$\mathbb{P}(J \geq t) \leq \mathbb{E} \min\{1, 2^{Z-t}\} = \mathbb{P}(Z + E \geq t),$$

where E is independent exponential of rate $\ln 2$. Integrating these nonnegative-variable tails yields $H(S_G) \leq L + 1/\ln 2$.

For the converse, under any deterministic policy each seed atom is contained in its transcript atom. Its surprisal therefore dominates the transcript surprisal under the induced coupling. Hence $F_S \leq F_{I_{\pi}}$ for all π , and thus $F_S \leq F_*$. Integration gives $H(S) \geq L$. This proves (6). $\square$

In the equal-entropy memoryless model, $F_*(t) = G_n(t - nh)$ by (4). Lemma 2 proves the first inequality in (5). In particular, realizability introduces no error proportional to the number of rounds or blocks.

3 An explicit finite-horizon Bellman comparison

Assume for now $a \geq b > 0$. Let Φ be the standard normal CDF and set

$$F(x) = \begin{cases} \frac{2b}{a+b} \Phi(x/b), & x \leq 0, \\ 1 - \frac{2a}{a+b} (1 - \Phi(x/a)), & x > 0. \end{cases} \quad (9)$$

This is a CDF. Define

$$K = \frac{\sqrt{2/\pi}}{a+b}, \quad m_3 = \max_A \mathbb{E}|D_A|^3, \quad \mathcal{C} = K \left(\frac{m_3}{3b^2} + a \right), \quad u_t(x) = F(x/\sqrt{t}) \quad (t > 0).$$

Lemma 3 (Explicit comparison). *For every $\tau \geq 1$, $R \geq 0$, and integer $n \geq 0$, put*

$$\eta = \max\{F(-R/\sqrt{\tau}), 1 - F(R/\sqrt{\tau})\}.$$

Then, for every $x \in \mathbb{R}$,

$$\boxed{\begin{aligned} F\left(\frac{x-R}{\sqrt{n+\tau}}\right) - \eta - \frac{3\mathcal{C}}{\sqrt{\tau}} &\leq G_n(x), \\ G_n(x) &\leq F\left(\frac{x+R}{\sqrt{n+\tau}}\right) + \eta + \frac{3\mathcal{C}}{\sqrt{\tau}}. \end{aligned}} \quad (10)$$

Proof. For x of either sign let $c = b$ on the negative half-line and $c = a$ on the positive half-line. Direct differentiation gives

$$F'(x) = Ke^{-x^2/(2c^2)}, \quad F''(x) = -\frac{Kx}{c^2}e^{-x^2/(2c^2)}.$$

The first two derivatives extend continuously across zero; in particular $F''(0) = 0$. The one-sided third derivatives satisfy

$$|F'''(x)| = \frac{K}{c^2} \left| \frac{x^2}{c^2} - 1 \right| e^{-x^2/(2c^2)} \leq \frac{2K}{b^2}.$$

Consequently F'' is globally Lipschitz with that constant. Taylor's theorem therefore applies even when an increment crosses the interface.

The function u_t satisfies the following identity everywhere, including $x = 0$:

$$\partial_t u_t(x) = \frac{1}{2} \min\{a^2 \partial_{xx} u_t(x), b^2 \partial_{xx} u_t(x)\}. \quad (11)$$

On $x < 0$ the second derivative is positive and the minimum selects b^2 ; on $x > 0$ it is negative and selects a^2 . On each side both expressions equal $-x \partial_x u_t(x)/(2t)$.

Because $\mathbb{E}D_A = 0$, the spatial Taylor remainder gives

$$|Tu_t - u_t - \partial_t u_t| \leq \frac{Km_3}{3b^2} t^{-3/2} \quad \text{uniformly in } x.$$

For clarity, the minimum of the two quadratic Taylor terms is exactly (11); taking a minimum cannot increase the maximal absolute remainder. Also, with $z = x/\sqrt{t}$,

$$\partial_{tt} u_t(x) = \frac{K}{4t^2} (3z - z^3/c^2) e^{-z^2/(2c^2)}, \quad \|\partial_{tt} u_t\|_\infty \leq \frac{2Ka}{t^2}.$$

The last bound follows from $\sup_{s \geq 0} se^{-s^2/2} \leq 1$ and $\sup_{s \geq 0} s^3 e^{-s^2/2} \leq 2$. The time Taylor remainder is thus at most Kat^{-2} . For $t \geq 1$,

$$\|Tu_t - u_{t+1}\|_\infty \leq Ct^{-3/2}. \quad (12)$$

The operator T is monotone, commutes with spatial translations and addition of constants, and is nonexpansive in the sup norm. Telescoping (12) therefore gives

$$\|T^n u_\tau - u_{n+\tau}\|_\infty \leq C \sum_{j=0}^{n-1} (\tau + j)^{-3/2} \leq \frac{3C}{\sqrt{\tau}}. \quad (13)$$

Finally, the definition of η implies the pointwise initial bounds

$$u_\tau(x - R) - \eta \leq \mathbf{1}_{\{x \geq 0\}} \leq u_\tau(x + R) + \eta.$$

Apply T^n , use its three invariances, and then (13). This proves (10). $\square$

Corollary 4 (CDF limit when $b > 0$). *For each $x \in \mathbb{R}$, $G_n(x\sqrt{n}) \rightarrow F(x)$.*

Proof. Fix τ, R and substitute $x\sqrt{n}$ in (10). Both shifted arguments tend to x , so the limsup and liminf differ from $F(x)$ by at most $\eta + 3C/\sqrt{\tau}$. First choose τ large and then R large. The error tends to zero. For explicit choices at tolerance $0 < \epsilon < 1$, take

$$\tau \geq \max\{1, (6C/\epsilon)^2\}, \quad R = a\sqrt{2\tau \ln(4/\epsilon)}.$$

The Gaussian Chernoff bound gives $\eta \leq \epsilon/2$, and the other term is at most $\epsilon/2$. These parameters are fixed before sending n to infinity, so no uniform convergence over a moving policy grid is being assumed. $\square$

4 Zero variance and convergence of the means

Lemma 5 (The degenerate endpoint). *If $a > 0$ and $b = 0$, then*

$$G_n(x\sqrt{n}) \longrightarrow F_{a,0}(x) = \begin{cases} 0, & x \leq 0, \\ 2\Phi(x/a) - 1, & x > 0. \end{cases}$$

Proof. This step concerns the centered increment control problem only. For $\varepsilon > 0$ add an independent Rademacher increment $\varepsilon\xi_t$ at each time and for either action. Retain both the original output and ξ_t as observed labels. The new centered increments have variances $a^2 + \varepsilon^2$ and ε^2 ; write G_n^ε for their Bellman CDF. Extra labels do not change the Bellman recursion: conditional on the history the next increment law depends only on the action, and the remaining terminal threshold depends only on the current sum.

The perturbed and unperturbed sums under the same action choices differ by $\varepsilon \sum_{t=1}^n \xi_t$. For every $d > 0$, Chebyshev gives

$$\mathbb{P}\left(\left|\varepsilon \sum_{t=1}^n \xi_t\right| > d\sqrt{n}\right) \leq \varepsilon^2/d^2,$$

uniformly over policies. Every original policy is a perturbed policy that ignores the extra coins. Conversely, an original controller can emulate a perturbed policy by drawing those coins as private randomness. The optimal original value is no larger than that randomized policy's value. These two observations give

$$G_n^\varepsilon((x-d)\sqrt{n}) - \varepsilon^2/d^2 \leq G_n(x\sqrt{n}) \leq G_n^\varepsilon((x+d)\sqrt{n}) + \varepsilon^2/d^2. \quad (14)$$

Apply Corollary 4 with $a_\varepsilon = \sqrt{a^2 + \varepsilon^2}$ and $b_\varepsilon = \varepsilon$. Then let $\varepsilon \downarrow 0$ with $d = \sqrt{\varepsilon}$. The explicit CDFs (9) converge to the continuous CDF $F_{a,0}$, also at the moving arguments $x \pm \sqrt{\varepsilon}$. At $x = 0$, both limits are zero directly from the two formulas. Equation (14) proves the claim. $\square$

Lemma 6 (Mean limit). *For $a \geq b \geq 0$,*

$$\lim_{n \rightarrow \infty} \frac{e_n}{\sqrt{n}} = \sqrt{\frac{2}{\pi}} (a - b). \quad (15)$$

Proof. Let $B = \max_{A,y} |-\log_2 P_A(y) - h|$. If $B = 0$ the claim is immediate; assume $B > 0$. For any policy the centered increments have conditional mean zero and absolute value at most B . Convexity bounds their conditional exponential moment by $\cosh(\lambda B) \leq e^{\lambda^2 B^2/2}$. Iteration and exponential Markov therefore give, uniformly over all policies,

$$\mathbb{P}(M_{\pi,n} > t\sqrt{n}), \quad \mathbb{P}(M_{\pi,n} \leq -t\sqrt{n}) \leq e^{-t^2/(2B^2)} \quad (t > 0).$$

By (4), both tails of $Z_n/\sqrt{n}$ have these same bounds. Hence the signed tail formula

$$\frac{e_n}{\sqrt{n}} = \int_0^\infty [1 - G_n(t\sqrt{n})] dt - \int_0^\infty G_n(-t\sqrt{n}) dt$$

admits dominated convergence, using Corollary 4 or Lemma 5. When $b > 0$, its limit is

$$\begin{aligned} \int_0^\infty [1 - F(t)] dt - \int_0^\infty F(-t) dt &= \frac{2a^2}{(a+b)\sqrt{2\pi}} - \frac{2b^2}{(a+b)\sqrt{2\pi}} \\ &= \sqrt{2/\pi} (a - b). \end{aligned}$$

Here $\int_0^\infty [1 - \Phi(t/c)] dt = c/\sqrt{2\pi}$. For $b = 0 < a$ the negative integral vanishes and the positive one is $a\sqrt{2/\pi}$. If $a = b = 0$, both increment laws are point masses at zero and $G_n = G_0$, so the assertion is immediate. $\square$

Proof of Theorem 1. Lemma 2 and Lemma 6 give

$$\frac{C_n^{\text{causal}} - nh}{\sqrt{n}} = \frac{e_n}{\sqrt{n}} + O(n^{-1/2}) \longrightarrow \sqrt{2/\pi} (a - b).$$

If both varentropies vanish, both laws are uniform. Equal entropy means their supports have the same size k . A single uniform seed on $\{1, \dots, k\}^n$, with the appropriate output relabeling for each requested action, has entropy $n \log_2 k = nh$. The all- P transcript gives the reverse inequality. The stronger bound (2) is proved in the following section. $\square$

5 Equal positive varentropies: the logarithmic upper bound

The main second-order argument above is self-contained. For the sharper remainder when $a = b = \sigma > 0$, we use the following specialization of Fan–Grama–Liu, Theorem 2.1 [4].

External input (with its exact hypotheses). Let $(\xi_i, \mathcal{F}_i)$ be martingale differences such that, almost surely,

$$\sum_{i=1}^n \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) = 1, \quad \left| \mathbb{E}(\xi_i^k \mid \mathcal{F}_{i-1}) \right| \leq \frac{1}{2} k! \epsilon^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) \quad (k \geq 2),$$

where $0 < \epsilon \leq 1/2$. There is an absolute constant C_{FGL} such that, for all real x ,

$$\left| \mathbb{P} \left(\sum_i \xi_i \leq x \right) - \Phi(x) \right| \leq C_{\text{FGL}} (1 + x^2) \epsilon |\ln \epsilon| \exp \left(-\frac{\hat{x}_\epsilon^2}{2} \right), \quad \hat{x}_\epsilon = \frac{2|x|}{1 + \sqrt{1 + 2\epsilon|x|}}. \quad (16)$$

This is that theorem with its quadratic-variation error parameter $\delta = 0$.

Proof of (2). Put $B = \max_{A,y} |-\log_2 P_A(y) - h|$. For every deterministic policy use

$$\xi_i = \frac{D_{A_i}(Y_i)}{\sigma \sqrt{n}}, \quad \epsilon_n = \frac{B}{\sigma \sqrt{n}}.$$

The conditional variance of every ξ_i is exactly $1/n$, regardless of which action the policy chooses. Also $|\xi_i| \leq \epsilon_n$, so

$$\left| \mathbb{E}(\xi_i^k \mid \mathcal{F}_{i-1}) \right| \leq \epsilon_n^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) \leq \frac{1}{2} k! \epsilon_n^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}).$$

Thus all the hypotheses hold once $n \geq 4B^2/\sigma^2$, with identical parameters for all policies. Taking the infimum of their CDFs preserves the same two-sided error about Φ . Consequently

$$|G_n(\sigma \sqrt{n} x) - \Phi(x)| \leq C_{\text{FGL}} (1 + x^2) \epsilon_n |\ln \epsilon_n| e^{-\hat{x}_{\epsilon_n}^2/2}. \quad (17)$$

This is a pointwise uniform bound, not merely a separate Wasserstein estimate for each policy.

The right side has an integrable envelope independent of $\epsilon_n \leq 1/2$. Indeed, for $|x| \geq 1$,

$$1 + \sqrt{1 + 2\epsilon_n|x|} \leq 1 + \sqrt{1 + |x|} \leq 3\sqrt{|x|}, \quad \hat{x}_{\epsilon_n}^2/2 \geq 2|x|/9.$$

It follows that

$$\int_{\mathbb{R}} (1 + x^2) e^{-\hat{x}_{\epsilon_n}^2/2} dx \leq \frac{8}{3} + 2 \int_0^\infty (1 + x^2) e^{-2x/9} dx < 400.$$

Since the standard normal has mean zero, the CDF formula for the difference of means and (17) give

$$\begin{aligned} 0 \leq e_n &\leq \sigma\sqrt{n} \int_{\mathbb{R}} |G_n(\sigma\sqrt{n}x) - \Phi(x)| dx \\ &\leq 400C_{\text{FGL}}B \ln\left(\frac{\sigma\sqrt{n}}{B}\right). \end{aligned}$$

The nonnegativity follows also from stochastic dominance over any single centered policy sum. Lemma 2 now gives, for every integer $n \geq 4B^2/\sigma^2$,

$$0 \leq C_n^{\text{causal}} - nh \leq 400C_{\text{FGL}}B \ln\left(\frac{\sigma\sqrt{n}}{B}\right) + \log_2 e.$$

Absorbing the finitely many smaller horizons into the constant proves (2). $\square$

6 The crash pair and the scope of the proof

For $P = (1/2, 1/4, 1/4)$ and $Q = (q, q, 1 - 2q)$ with $q \in (1/3, 1/2)$ defined by $H(Q) = 3/2$, one obtains

$$\boxed{C_n^{\text{causal}}(P, Q) = \frac{3}{2}n + 0.034747254051818\dots\sqrt{n} + o(\sqrt{n}).} \quad (18)$$

The exact coefficient is $\sqrt{2/\pi}(1/2 - \sqrt{V_Q})$, where

$$V_Q = 2q(1 - 2q) \left(\log_2 \frac{q}{1 - 2q} \right)^2.$$

The exact-rational certificate in Section 8 encloses the root, variance, and coefficient with explicit analytic remainders. The rounded display in (18) is not the definition of the coefficient.

What is repaired relative to the supplied draft. The proof uses the envelope over all adaptive transcript laws itself. There is no need to identify a seed-dependent joint diffusion limit. The discrete identity (11) has a minimum, and its sign is verified on both half-lines. The finite-horizon inequality (10) replaces the unquantified diffusion/grid passage. Finally, the single charge (8) replaces the unproved fiber-recycling estimate: it never accumulates a coder loss per block.

The draft's additional assertion $C_n^{\text{causal}} = nh + O(\log n)$ when $a = b > 0$ does not follow from a vanishing square-root coefficient or a fourth-moment bound. Section 5 proves it instead by an explicitly stated external theorem, with all its hypotheses checked uniformly in the policy. The full second-order formula (1) and the stated logarithmic upper bound are therefore both established. This does not assert a sharp logarithmic coefficient or a logarithmic lower bound.

Verification and computational scope. The included exact verifier checks all residual flow identities, reconstructed decoder marginals, and soft-tail inequalities in eighteen finite instances, including history-dependent kernels and three actions. It also includes a depth-three decoder certificate. The separate Bellman-limit program provides floating-point illustrations for $(a, b) = (2, 1)$ and $(1, 0)$; these numerical runs are not used in the proof. The universal argument is the finite-tree proof and the explicit inequalities above (and the stated external theorem for the equal-variance refinement), not an extrapolation of tests.

The source and decoder may be exponential in n . There is no claim of polynomial-time simulation, of one infinite-horizon seed, or of a proof-assistant certificate. Literature priority has not been established. Ordinary one-round coupling bounds and information-spectrum converses are prior work [1, 2, 3]; the response-tree realization argument needed here is given in full in Lemma 2.

7 Reconciling the four audit requests

Lower bound. The threshold-policy-grid route in the earlier draft is replaced, not completed by an implicit appeal to diffusion convergence. Equation (4) optimizes over every deterministic adaptive policy at each finite horizon. Lemma 3 compares that exact finite-horizon value with the explicit profile (9). Lemma 6 computes its mean and supplies a policy-uniform integrable tail bound. Thus the mean limit does not rely on convergence of a growing grid, or on interchanging a joint diffusion limit with an infimum. Section 8 makes its finite- n error explicit for the crash pair.

Upper bound. The block/recycling route is likewise replaced. Lemma 2 constructs a single exact response-tree seed and bounds its entire entropy overhead by $\log_2 e$. No rounding of an arithmetic coordinate and no antidiagonal lift are performed. In particular, charging a constant per block of length $(\log n)^2$ would only give $O(n/(\log n)^2)$, which is not $o(\sqrt{n})$; the original route cannot be certified by such an accounting argument. The global residual-flow identity (7) is the exactness certificate for the replacement.

Which profile gives one half? For the two constant policies alone, the centered normalized transcript CDFs converge, when $a \geq b > 0$, to $\Phi(x/a)$ and $\Phi(x/b)$. Their lower envelope is

$$F_{\text{fixed}}(x) = \begin{cases} \Phi(x/b), & x \leq 0, \\ \Phi(x/a), & x > 0. \end{cases}$$

The same uniform Gaussian tail argument gives its mean

$$\int_0^\infty (1 - \Phi(t/a)) dt - \int_0^\infty \Phi(-t/b) dt = \frac{a - b}{\sqrt{2\pi}}.$$

This is half of (15). The full adaptive-policy envelope is G_n , not the envelope of those two policies. Its limit is (9) and has twice this mean. Thus a statement that *every* policy-profile argument gives only half the coefficient would be false in the model defined in Section 1.

The quoted assertion of the original “Theorem 7”, namely that marginal transcripts cannot yield more than $O(\sqrt{n})$, does not contradict a sharp positive multiple of $\sqrt{n}$. The present proof identifies the coefficient within that order, and adds an achievability argument using the compatible history tree. This comparison addresses only the quoted assertion: the complete text and hypotheses of that Theorem 7 have not been supplied with this audit, so no stronger statement about its content is made.

What the numerical audit does. The following table encloses the *actual optimum* using a proved finite- n inequality. It is not a direct numerical optimization of C_n at those horizons, nor a Bellman discretization passed off as an exact seed cost. The executable checks the constants and endpoint rounding using rational arithmetic. The analytic inequalities remain an ordinary mathematical proof, not a proof-assistant certificate.

8 An explicit error bound for the crash pair

Let q be the unique root in $(1/3, 1/2)$ of

$$-2q \log_2 q - (1 - 2q) \log_2(1 - 2q) = 3/2.$$

Uniqueness follows since the derivative is $2 \log_2((1 - 2q)/q) < 0$ on this interval, and the endpoint entropies are $\log_2 3$ and 1. Put

$$a = \frac{1}{2}, \quad b = \sqrt{2q(1 - 2q)} \log_2 \frac{q}{1 - 2q}, \quad c = \sqrt{2/\pi} (a - b).$$

These are exact definitions. The accompanying rational computation gives

$$\begin{aligned} 0.410032428181985576080116712465 &< q < 0.410032428181985576080116712466, \\ 0.456450775263963373017231101124 &< b < 0.456450775263963373017231101125, \\ 0.034747254051817864080022137947 &< c < 0.034747254051817864080022137948. \end{aligned}$$

Theorem 7 (Quantitative crash-pair certificate). *For every integer $n \geq 16$,*

$$\left| \frac{C_n^{\text{causal}} - 3n/2}{\sqrt{n}} - c \right| \leq \frac{17}{2} \frac{\sqrt{\ln n}}{n^{1/4}} + \frac{3}{\sqrt{n \ln n}} + \frac{2}{\sqrt{n}}. \quad (19)$$

In particular the unnormalized remainder is $O(n^{1/4}\sqrt{\ln n})$, with the displayed explicit bound.

Proof. The rational certificate verifies $9/20 < b < 1/2$ and $B < 1$, where B is the increment bound in Lemma 6. Since $|D_A|^3 \leq BD_A^2$, it follows that $m_3 \leq 1/4$. Also $\pi > 2$, so $K \leq 20/19$ and

$$\mathcal{C} \leq \frac{20}{19} \left(\frac{1/4}{3(9/20)^2} + \frac{1}{2} \right) = \frac{4430}{4617} < 1.$$

For a given $n \geq 16$ choose

$$\tau = \sqrt{n}, \quad \eta_0 = n^{-1/4}, \quad R = a\sqrt{2\tau \ln(2/\eta_0)}, \quad s = \sqrt{1 + n^{-1/2}}, \quad r = R/\sqrt{n}, \quad T = \sqrt{\ln n}.$$

The Gaussian Chernoff bound and the prefactors at most two in (9) give $\eta \leq \eta_0$. Thus, with $d = \eta + 3\mathcal{C}/\sqrt{\tau} \leq 4n^{-1/4}$, Lemma 3 reads

$$F((x - r)/s) - d \leq G_n(x\sqrt{n}) \leq F((x + r)/s) + d.$$

Let $Y = Z_n/\sqrt{n}$ and let W have CDF F , so $\mathbb{E}W = c$. Integrating the last comparison over $[-T, T]$ gives

$$\left| \int_{-T}^T [F(x/s) - G_n(x\sqrt{n})] dx \right| \leq r + 2Td. \quad (20)$$

To see the constant r explicitly, for any CDF H , $\int_{\mathbb{R}} [H(x) - H(x - r)] dx = r$ when $r \geq 0$. Apply this to $H(x) = F(x/s)$ for the upper signed bound and to $H(x + r) - H(x)$ for the lower one. Each integrand is nonnegative, so restricting its integral to $[-T, T]$ can only decrease it.

For $t > 0$, Lemma 6 and (9) give respectively

$$\mathbb{P}(|Y| > t) \leq 2e^{-t^2/(2B^2)}, \quad \mathbb{P}(|sW| > t) \leq 2e^{-t^2/(2a^2s^2)}.$$

The latter follows by bounding both Gaussian scales by a ; their prefactors sum to two. For any $v > 0$, $\int_T^\infty e^{-t^2/(2v)} dt \leq (v/T)e^{-T^2/(2v)}$. Since $B^2 \leq 1$, $a^2s^2 \leq 1$, and $2B^2 + 2a^2s^2 \leq 3$, the total contribution outside $[-T, T]$ to the absolute signed difference of the means is at most

$$\frac{3}{T} e^{-T^2/2} = \frac{3}{\sqrt{n \ln n}}.$$

Consequently (20) bounds $|\mathbb{E}Y - sc|$ by $r + 2Td + 3/\sqrt{n \ln n}$. Next, $0 \leq (s - 1)c \leq 1/(2\sqrt{n})$, using $c < 1$ and $\sqrt{1 + u} - 1 \leq u/2$. For $n \geq 16$, $\ln 2 \leq \frac{1}{4} \ln n$, whence

$$r = \frac{1}{2} n^{-1/4} \sqrt{2 \ln(2n^{1/4})} \leq \frac{1}{2} n^{-1/4} \sqrt{\ln n}, \quad 2Td \leq 8n^{-1/4} \sqrt{\ln n}.$$

Finally, Lemma 2 changes $\mathbb{E}Y$ to $(C_n^{\text{causal}} - 3n/2)/\sqrt{n}$ by at most $\log_2 e/\sqrt{n} < 3/(2\sqrt{n})$; the last inequality follows from $\ln 2 > 2/3$. Adding these bounds proves (19). $\square$

n	Certified lower endpoint	Certified upper endpoint
10^8	0.000000000000	0.399831377112
10^{12}	0.000000000000	0.079430259815
10^{16}	0.029587976869	0.039906531235
10^{20}	0.034170431871	0.035324076233
10^{24}	0.034684066372	0.034810441732
10^{32}	0.034746524423	0.034747983681
10^{48}	0.034747253962	0.034747254142

Every row encloses $(C_n^{\text{causal}} - 3n/2)/\sqrt{n}$. Endpoints are rounded outward to twelve decimal places. Nonnegativity is also used for the first two lower endpoints. Already the row $n = 10^{16}$ excludes $c/2 = 0.0173736270259089\dots$; later rows narrow around c .

Reproduction and arithmetic certificate. Run `python3 certify_crash_pair.py`. Only the Python standard library is required. The output JSON records the coefficient intervals, all table rows, and the status `PASS_EXACT_RATIONAL_ARITHMETIC`. Logarithms are enclosed with 64 terms of the positive series

$$\ln x = 2 \sum_{j=0}^{63} \frac{z^{2j+1}}{2j+1} + R, \quad z = \frac{x-1}{x+1}, \quad 0 \leq R \leq \frac{2z^{129}}{129(1-z^2)} \quad (1 \leq x \leq 2),$$

after powers-of-two range reduction. Entropy signs are checked exactly at both endpoints at every root-bracketing stage. Square roots use integer-square-root enclosures; π uses Machin’s identity with alternating-series remainders. Every decimal enclosure is rounded outward. No floating-point value is accepted as an arithmetic premise. The script certifies this arithmetic, not the universal analytic theorem by formal verification. The standalone proof above supplies that theorem.

References

- [1] S. Compton. *A Tighter Approximation Guarantee for Greedy Minimum Entropy Coupling*. 2022. [arXiv:2203.05108](#).
- [2] S. Compton, D. Katz, B. Qi, K. Greenewald, and M. Kocaoglu. *Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier*. AISTATS, PMLR 206, 10445–10469, 2023. [Proceedings](#).
- [3] Y. Y. Shkel and A. K. Yadav. *Information Spectrum Converse for Minimum Entropy Couplings and Functional Representations*. ISIT, 2023. [arXiv:2305.05745](#).
- [4] X. Fan, I. Grama, and Q. Liu. *Nonuniform Berry–Esseen bounds for martingales with applications to statistical estimation*. Theorem 2.1, conditions (A1)–(A2). [arXiv:1608.05217](#).